MTH355

Timed Online Assignment – January Semester 2020

Basic Mathematical Optimization

Tuesday, 12 May 2020 **1:00 pm – 3:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) comprises **FIVE (5)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, while the assignment is in session, briefly explain your understanding of and assumptions about that question before attempting it.

4. You are to include the following particulars in your submission: Course Code, Full Name and Student PI and name your submission file as - CourseCode_FullName_StudentPI.

5. For answers which are hand-written, ensure that the question number is clearly stated on each page. All uploaded hand written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

6. Please submit only ONE (1) file (<500 MB) in either **PDF/JPEG/WORD format** within the time-limit via Canvas [similar to Tutor Marked Assignment (TMA) submission]. If you do not submit within the time-limit, you would be deemed to have withdrawn (W) from the course. Appeal is **NOT** allowed.

7. To prevent plagiarism and collusion, your submission will be reviewed thoroughly by Turnitin, The Turnitin report will only be made available to the markers. The university takes a tough stance against plagiarism and collusion. Serious cases will normally result in the student being referred to SUSS's Student Disciplinary Group. For other cases, significant marking penalties or expulsion from the course will be imposed.

Answer all questions. (Total 100 marks)

Question 1

(a) Consider the following system of linear equations.

$$\begin{aligned} 2x_1 + 3x_2 + x_3 &= 2 \\ 4x_1 + 3x_2 - x_3 &= -1 \\ 2x_1 + 3x_2 &= 0 \end{aligned}$$

(i) Write down the matrix A and vector b, where $Ax = b$. Apply the LU decomposition technique on A. Find the elementary matrices E_{ij}.

(7 marks)

(ii) Based on the LU of A in Question 1(a)(i), solve the system of linear equations.

(5 marks)

(iii) Use the LU of A to find the second column of A^{-1}.

(3 marks)

(b) Consider the system of linear equations.

$$\begin{aligned} 3x_1 - 5x_2 + x_3 &= 3 \\ 7x_1 + 3x_2 + 2x_3 &= 2 \\ -4x_1 + 3x_2 + 10x_3 &= -3 \end{aligned}$$

(i) Apply the conditions of convergence for a set of linear equations. Starting from the point$(x_1, x_2, x_3) = (1,2,3)$, perform three iterations of the Gauss-Seidel iterative scheme. Keep all values up to five decimal places.

(6 marks)

(ii) Starting from the point $(x_1, x_2, x_3) = (3,2,1)$, perform three iterations of the Jacobi iterative scheme. Keep all values up to five decimal places.

(4 marks)

Question 2

(a) Solve the following linear optimization problem with the two-phase simplex solution technique.

$$\begin{aligned} \text{Maximize } z &= \frac{x_1}{4} + x_2 \\ x_1 + 2x_2 &\leq 2 \\ 3x_1 + x_2 &= 3 \\ x_1, x_2 &\geq 0 \end{aligned}$$

(13 marks)

(b) Consider the following linear programming problem.

$$\begin{aligned} \text{Maximize } z &= 2x_1 + x_2 + 4x_3 \\ 5x_1 + 3x_2 + 4x_3 &\leq b_1 \\ 3x_1 + 2x_2 + 4x_3 &\leq b_2 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

(i) Solve the linear optimization modelling problem using the simplex method when $b_1 = 40$ and $b_2 = 25$.

(6 marks)

(ii) When $b_1 = 20$ and $b_2 = 40$, revise the optimal tableau obtained in Question 2(b)(i) to find the new optimal solution for the new linear programming problem.

(6 marks)

Question 3

(a) Solve the following linear optimization problem using the big-M method.

$$\begin{aligned} \text{Maximize } z &= 1104x_1 + 138x_2 \\ 4x_1 + x_2 &= 4 \\ x_1 + 6x_2 &\geq 6 \\ x_1, x_2 &\geq 0 \end{aligned}$$

(9 marks)

(b) Consider the following primal linear programming problem.

$$\begin{aligned} \text{Minimize } z &= 3x_1 + 4x_2 + 5x_3 \\ 2x_1 + x_2 + x_3 &\geq 1 \\ x_1 + 4x_2 + 2x_3 &\geq 2 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

(i) Determine the dual model.

(4 marks)

(ii) If the optimal solution of the primal model is $(x_1, x_2, x_3) = (\frac{2}{7}, \frac{3}{7}, 0)$, use the complementary slackness optimality conditions to compute the optimal solution of the dual model.

(4 marks)

(c) If a linear programming problem is completely solvable by the simplex method, give one possible way to deduce the number of basic variables.

(4 marks)

(d) Could a linear programming problem possess more than one optimal solution? If yes, give a simple example.

(4 marks)

Question 4

(a) Winsor Co has five potential projects to choose from for the next 2-year period. The yearly expenditures in millions dollars ($M) and returns ($M) for each project are tabulated in Table Q4(a) below.

Project	Year 1 ($M)	Year 2 ($M)	Return ($M)
A	35	41	150
B	18	20	100
C	35	32	140
D	22	23	86
E	20	18	75

Table Q4(a)

The available funds for the first and second years are $120M and $140M, respectively. The company can take on a maximum of 4 projects. In order to maintain the existing workforce, it must engage in a minimum of 2 projects. The company has to decide which projects should be executed over the 2-year period.

Formulate the linear optimization problem above as a 0-1 variable model where the objective is to maximize the profit. Determine the constraints involved.

(8 marks)

(b) Consider the following integer linear programming model.

$$\begin{aligned} \text{Maximize } z &= 6x_1 + 4.5x_2 \\ 2x_1 + x_2 &\leq 9 \\ 5x_1 + 9x_2 &\leq 54 \\ x_1, x_2 &\geq 0 \end{aligned}$$

x_1, x_2 are integers.

(i) Determine graphically, the solution to the continuous problem.

(5 marks)

(ii) Employ the branch and bound method to solve the integer linear programming model. Draw the tree diagram for the solution process. Construct a table of results using the following format.

Problem, I_k	Problem solved, C_k	Solution to C_k	Current bound	Problem to pursue	Stored problem

(12 marks)

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